Academic Data Module

Entity Tables

Institution

Attribute	Constraints
Id	Unique (PK)
name	Unique
created_at	Required
updated_at	Required

Department

Attribute	Constraints
Id	Unique (PK)
institution_id	Required, FK
name	Required
code	Unique
description	Optional
created_at	Required
updated_at	Required

Admin

Attribute	Constraints
Id	Unique (PK)
institution_id	Required, FK
user_id	Optional, FK
first_name	Required
last_name	Required
created_at	Required
updated_at	Required

Faculty Member

Attribute	Constraints
Id	Unique (PK)
user_id	Optional, FK
department_id	Required, FK
first_name	Required
last_name	Required
role	Required

access_code	Unique
date_of_joining	Required
created_at	Required
updated_at	Required

Student

Attribute	**Constraints**
Id	Unique (PK)
first_name	Required
last_name	Required
department_id	Required, FK
degree_id	Required, FK
registration_number	Unique
enrollment_date	Required
academic_status	Required
semester	Required
created_at	Required
updated_at	Required

Degree

Attribute	**Constraints**
Id	Unique (PK)
title	Unique
code	Required
department_id	Required, FK
created_at	Required
updated_at	Required

Session

Attribute	**Constraints**
Id	Unique (PK)
name	Required
start_date	Required
end_date	Optional
is_expired	Derived (from end_date)
created_at	Required
updated_at	Required

Course

Attribute	Constraints
Id	Unique (PK)
degree_id	Required, FK
title	Required
code	Optional, Unique
semester	Required
credit_hours	Optional
created_at	Required
updated_at	Required

Course Enrollment

Attribute	Constraints
Id	Unique (PK)
course_id	Required, FK
student_id	Required, FK
session_id	Required, FK
status	Required
created_at	Required
updated_at	Required

Course Assignment

Attribute	Constraints
Id	Unique (PK)
course_id	Required, FK
faculty_id	Required, FK
session_id	Required, FK
is_active	Required
created_at	Required
updated_at	Required

Exam

Attribute	Constraints
Id	Unique (PK)
course_id	Required, FK
session_id	Required, FK
title	Required
type	Required

status	Required
date_of_conduct	Required
max_marks	Required
created_at	Required
updated_at	Required

Exam Result

Attribute	**Constraints**
Id	Unique (PK)
student_id	Required, FK
exam_id	Required, FK
obtained_marks	Required
grade	Required
gpa	Optional
status	Required
created_at	Required
updated_at	Required

Attendance

Attribute	**Constraints**
Id	Unique (PK)
student_id	Required, FK
session_id	Required, FK
faculty_id	Required, FK
course_id	Required, FK
status	Required
date	Required
created_at	Required
updated_at	Required

Exam Log

Attribute	**Constraints**
Id	Unique (PK)
exam_id	Required, FK
faculty_id	Required, FK
action	Required
description	Optional
created_at	Required

updated_at	Required

Academic Event

Attribute	Constraints
Id	Unique (PK)
institution_id	Required, FK
course_id	Required, FK
session_id	Required, FK
source_id	Required, FK
type	Required
title	Required
description	Optional
created_at	Required
updated_at	Required

Relationship Tables

Institution

Entity	Type	Name
Department	1-M	Contains
Admin	1-1	Has
Faculty	1-M	Has
Student	1-M	Has

Department

Entity	Type	Name
Faculty	1-M	Has
Student	1-M	Has
Degree	1-M	Offers

Admin

Entity	Type	Name
Student	1-M	Registers
Faculty	1-M	Registers
Academic Event	1-M	Monitors
Exam Log	1-M	Analyzes
Session	1-M	Plans / Creates

Faculty Member

Entity	Type	Name

Course	1-M	Teaches
Course Assignment	1-M	Assigned
Exam	1-M	Conducts
Exam Result	1-M	Announces
Attendance	1-M	Marks
Academic Event	1-M	Triggers

Student

Entity	Type	Name
Degree	1-1	Enrolled
Course	1-M	Studies
Course Enrollment	1-M	Enrolled
Exam	1-M	Attends
Exam Result	1-M	Receives

Degree

Entity	Type	Name
Course	1-M	Offers

Session

Entity	Type	Name
Exam	1-M	Conducted
Attendance	1-M	Records
Course Assignment	1-M	Records
Course Enrollment	1-M	Records
Academic Event	1-M	Records

Course

Entity	Type	Name
Course Enrollment	1-M	Has
Course Assignment	1-1	Has
Exam	1-M	Has
Exam Result	1-M	Contains
Academic Event	1-M	Has

Exam

Entity	Type	Name
Exam Log	1-M	Contains